Nirma University

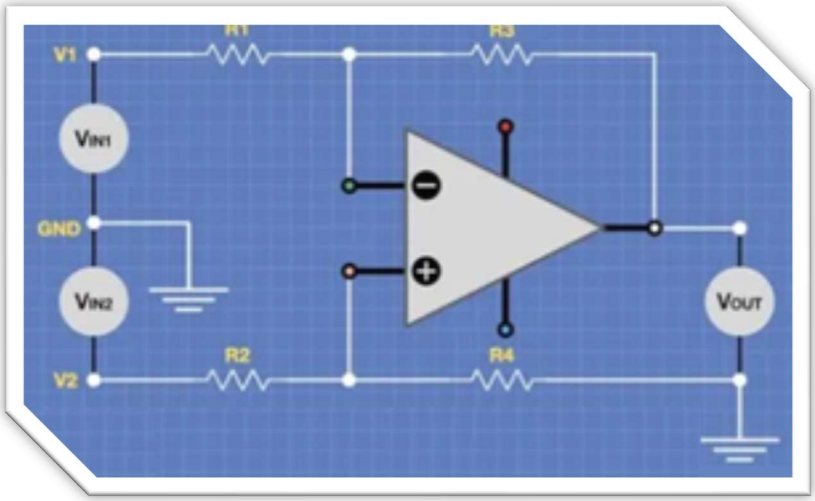
School of Technology

B. Tech. Sem. VII (EC)

4EC601ME25 Analog CMOS Integrated Circuits

Laboratory Manual

ODD 2026



NIRMA UNIVERSITY
INSTITUTE OF TECHNOLOGY
NAAC ACCREDITED 'A+' GRADE

Electronics & Communication Engineering Department,
School of Technology,
Ahmedabad

Course Learning Outcomes (CLOs)

At the end of the course, the students shall be able to:

1. apply noise analysis for analog CMOS circuits and evaluate input signal noise and output signal noise (BL3)
2. analyse the given analog circuit using a large signal, small signal and high- frequency models and evaluate performance parameters (BL4)
3. design an analog signal conditioning circuit using the operational amplifier for the given specific application (BL6)
4. construct an amplifier using switching capacitors circuits for the given specifications (BL6)

Index

SR NO	Title	Date	Marks as per Rubrics					Total Marks	Sign
			02	02	02	02	02		
1.	Exploring Cadence Virtuoso for ASIC Design **(CO1, CO2, CO3, CO4)**							**/10**	
2.	To simulate the output and transfer characteristics of MOS transistors for finding out technology parameters **(CO2)**							**/10**	
3.	To design and simulate current mirror circuits. **(CO2)**							**/10**	
4.	To Design and simulate MOS based resistive and current source load single stage CS amplifiers for given specifications. **(CO2, CO4)**							**/10**	
5.	To Design and simulate MOS based source degenerated single stage CS amplifiers for given specifications. **(CO2, CO4)**							**/10**	
6.	To Design and simulate MOS based single stage CD and CG amplifiers for given specifications. **(CO2, CO4)**							**/10**	
7.	To Design and simulate MOS based cascode amplifiers for given specifications. **(CO2, CO4)**							**/10**	
8.	To Design and simulate differential amplifiers for given specifications. **(CO2, CO4)**							**/10**	
9.	To design and simulate CMOS two stage operational Amplifier. **(CO3)**							**/10**	
10.	To carry out project work for the assigned topic. **(CO1, CO2, CO3, CO4)**							**/20**	

Roll No. **Date:**

Exp. 1. Exploring Cadence Virtuoso for ASIC Design

Learning Outcomes

After completion of this experiment students will be able to:

1. Utilize Cadence Virtuoso for Circuit Simulations
2. Carry out AC, DC and transient analysis
3. Prepare layout of circuit
4. Carry out pre and post layout simulations.

Pre-lab Exercise

1. List various tools available under Cadence VLSI design suite with their application area.
2. List VLSI design suites other than Cadence. Also list the tools under all suites with their application areas.

Procedure (Attach a separate sheet with all calculations):

1. Refer to the Cadence ASIC Design Manual sent to you by a separate email. The same is available on LMS.
2. Complete the Pre layout simulations and find out the threshold voltage and propagation delay for the CMOS inverter.
3. Prepare the layout and carry out DRC, LVS and RC extraction. Observe the various parasitic created after the layout.
4. Carry out post layout simulations and compare the threshold voltage and propagation delay results of pre and post layout simulations.

Conclusion:

Roll No. **Date:**

Exp. 2. To simulate the output and transfer characteristics of MOS transistors for finding out technology parameters.

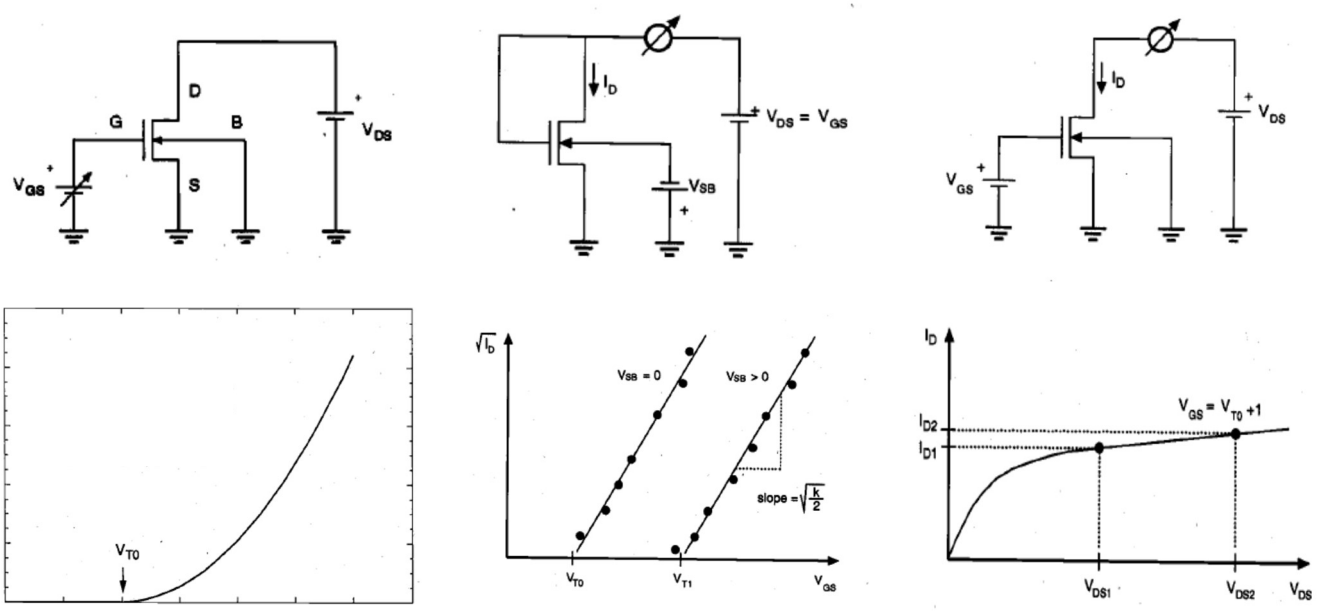
Learning Outcomes

After completion of this experiment students will be able to:

1. Prepare the schematic to find out the MOS characteristics
2. Carry out AC, DC and transient analysis
3. Determine the MOS operating regions depending on applied voltages
4. Extract basic parameters through experimental measurements

Pre – Lab Exercise

1. What are the different operating regions of MOSFET and conditions for the same?
2. Write down the equations of g_m, λ and γ. Also discuss the significance of the same.



A B C

Figure 1

Procedure

1. Prepare the designs for the circuits of Figure 1.
2. Figure 1 – A is Drain Current vs Gate to Source Voltage setup and expected graph. Simulate the same and from the graph find out the value of V_{T0}.
3. Figure 1 – B is Root of Drain Current vs Gate to Source Voltage setup and expected graph. Mind the root of current and act accordingly. The slope of the curve will provide Kn values. Change the value of Body potential and replot the same graph to find the value of γ. γ can be found out by changing body potential in the step 1 as well.

4. Figure 1 – C is Drain Current vs Drain to Source Voltage setup and expected graph. Simulate the same and from the graph find out the value of λ.
5. Take help from the text book: CMOS Digital Integrated Circuits: Analysis and Design by Sung – Mo – Kang and Yusuf Leblebici. Also write the equations to find out the theoretical values of the above listed parameters.

Assignment:

1. Repeat the experiment for PMOS device.

Conclusion:

Assignment:

1. Which transistor has maximum operating frequency, NMOS or PMOS? Justify your answer with proper reasoning. Support your answers with graphs if possible. What is the parameter that allows to find out the maximum operating frequency as transistor can work with?
2. Can body of the MOS be used as an input terminal? List the pros and cons of using the same as input terminal.

Roll No. **Date:**

Exp. 3. To design and simulate current mirror circuits.

Learning Outcomes:

After completion of this experiment students will be able to:

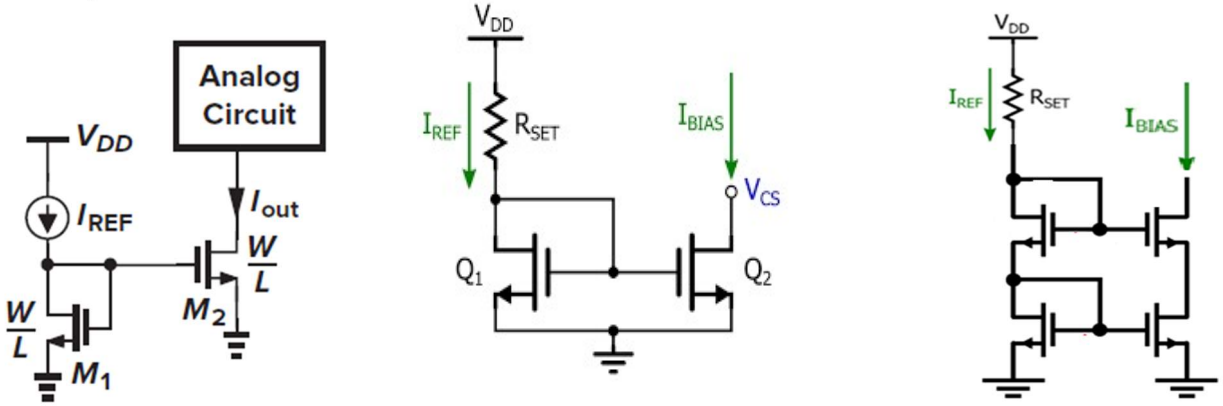
1. Introduce and characterize the current mirrors.
2. Design of different types of current mirrors.
3. Improve the performance of the current mirrors

Pre-lab Exercise:

1. Justify the need of biasing circuits. List the different biasing methods for amplifier design.
2. Discuss the limitations of the traditional biasing methods listed above.

Introduction:

A current mirror is a circuit designed to copy a current through one active device by controlling the current in another active device of a circuit, keeping the output current constant regardless of loading. The current being 'copied' can be, and sometimes is, a varying signal current. Conceptually, an ideal current mirror is simply an ideal *inverting current amplifier* that reverses the current direction as well or it is a current-controlled current source (CCCS). The current mirror is used to provide bias currents and active loads to circuits. It can also be used to model a more realistic current source (since ideal current sources don't exist).



A. Simple B. Experimental Setup C. Cascode

Figure 1. Current Mirror

Specifications:

Design a current mirror using V_{DD} = 1.8 V to sink a current of 10 uA.

Procedure:

1. Design schematic as shown in figure 1 B. At the V_{CS} point connect the resistor of the same value as that of R_{SET}. Vary the value of resistor at point V_{CS} and observe the current in both the branches.
2. Find out the range of values of resistor for which the change in current I_{BIAS} is less than 10% compared to the I_{REF}.

3. Now replace the convert the circuit into the Cascode Current mirror and repeat the steps 1 and 2.
4. Convert the R_{SET} to PMOS and design the circuit to get the current as per the specifications provided.
5. Compare the simulated result with specifications.

Results:

Range of Resistance In Simple Current Mirror	
Range of Resistance In Cascode Current Mirror	
Range of Resistance with PMOS	

Assignment:

1. Sketch small signal model of cascade current mirror. Compare theoretic and practical value of output resistance.

Conclusion:

Roll No. **Date:**

Exp. 4. To Design and simulate MOS based resistive and current source load single stage CS amplifiers for given specifications.

Learning Outcomes

1. After completion of this experiment students will be able to:
2. Find out the operating point of MOS device.
3. Bias the MOS device for appropriate operating region.
4. Implement resistive and current source load CS amplifier for given specifications.
5. Carryout AC, DC and Transient analysis for amplifier

Pre-lab Exercise

1. Draw the small signal model for the MOSFET. Describe the concept of the small signal and small signal model.
2. List various performance parameters for amplifier and explain them with equations.
3. Derive the equation for input impedance for Resistive Load Single Stage CS MOS based amplifier.

Introduction

Amplifier is one of the most important analog circuit that amplifies the input AC signal. The main aim of amplifier is to amplify the signal, but the other features like bandwidth, amount of noise added by the circuit, total harmonics introduced in the output etc. are also very important in the design of an amplifier. The simplest amplifier is a single stage resistive load amplifier. The following figure shows the CS single stage amplifier.

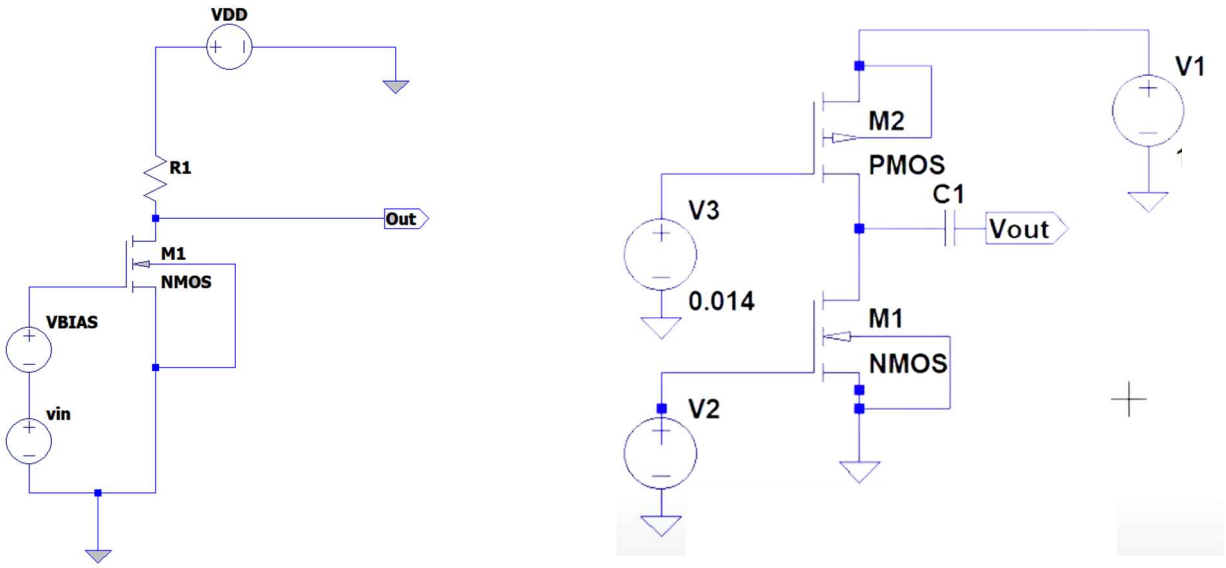


Figure 1. Single Stage CS Amplifier

Specification:

Gain	20 dB	Power	< 1mW

Procedure (Attach a separate sheet with all calculations):

1. For resistive load amplifier, Calculate the load resistance R_1 and current I_{DS} for given power requirements of the amplifier.
2. Find out the transconductance of the transistor required to achieve the specified gain.
3. Find out the value of C_{OX} using the model file.
4. Using the relation of g_m and I, find out W/L ratio and set the value of W and L.
5. Prepare the circuit for simulation as per the values found out.
6. Carry out AC, DC and Transient analysis for the amplifier.
7. Compare theoretical results with the experimental results.
8. Repeat the experiment with Current source Load first using NMOS as a load and replacing the same with PMOS as a load.
9. In step no. 8 no need to re calculate the values from beginning. For the available W/L of NMOS driver set the value of load W/L to achieve the same gain.

Results:

Amplifier Parameters			
	Resistive Load	**NMOS Load**	**PMOS Load**
Gain			
Bandwidth			
Noise			
Power			
Input AC voltage range			
Output AC voltage range			
MOSFET Parameters			
W driver			
W load			
gm			
Ron			
μ_n			
ro			
V_{T0}			
C_{GS}			
I_{DS}			
C_{OX}			

Assignment:

1. What is the maximum ac voltage that can be applied to the amplifier for proper output operation?

2. What is the maximum output swing that can be achieved? Write the upper and lower value of output swing.
3. List the effect of input DC value, Value of Resistor and Value of W on the gain, output swing and power of the amplifier.
4. What is the practical way to find out the operating point of the amplifier?

Conclusion:

Roll No. **Date:**

Exp. 5. To Design and simulate MOS based source degenerated single stage CS amplifiers for given specifications.

Learning Outcomes

1. Implement source degenerated CS amplifier for given specifications.
2. Carryout AC, DC and Transient analysis for amplifier
3. Carryout parametric analysis in Cadence Virtuoso.

Pre-lab Exercise

1. Common source amplifier suffers from non-linearity. Justify the statement.
2. Derive the equation for output resistance for source degenerated CS amplifier.

Introduction

To make the traditional common source amplifier a linear amplifier, a resistor is placed at the source of the driver transistor. This resistor weakens the dependence of gain of amplifier on the transconductance of the amplifier and makes it a linear amplifier. Figure 1 shows the source degenerated CS amplifier.

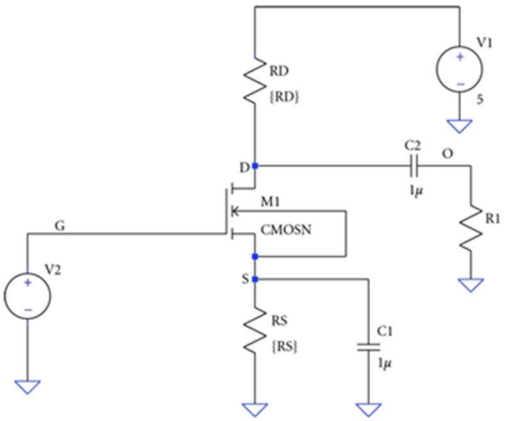


Figure 1. Source Degenerated Single Stage CS Amplifier

Procedure (Attach a separate sheet with all calculations):

1. Use the parameter values found in the last experiment.
2. Put a resistor at the source of the driver transistor as shown in figure 1.
3. Find the value of resistor to avail the required gain with existing parameters.
4. Carry out AC, DC and Transient analysis for the amplifier.
5. Compare theoretical results with the experimental results.
6. Find out the optimum value of resistor that provides the required gain using parametric analysis.

Results:

Amplifier Parameters			
Gain			
Bandwidth			
Noise			
Power			
Input AC voltage range			
Output AC voltage range			
MOSFET Parameters			
W driver		ro	
Rs		V_{T0}	
Gm		C_{GS}	
Ron		I_{DS}	
μ_n		C_{OX}	

Assignment:

1. What will be the effect on the gain of the amplifier if the source resistance is replaced by the transistor? Consider the case of both NMOS and PMOS transistor. Also comment on the issues or challenges encounter in the said replacement.

Conclusion:

Roll No. **Date:**

Exp. 6. To Design and simulate MOS based single stage CD and CG amplifiers for given specifications

Learning Outcomes

After completion of this experiment students will be able to:

1. Design a single stage CD and CG amplifier for given specifications.
2. Simulate CD and CG amplifier circuit and find out performance parameters of amplifier.

Pre-lab Exercise

1. Draw the small signal model for CG and CD amplifier.
2. Derive the equation for input impedance and gain for Single Stage CG and CD MOS based amplifier.
3. Justify the need of CD amplifier.

Introduction

After the study of CS stage, it is necessary to study the other alternatives in the amplifier. The first alternative is common gate. Figure 1 shows the CG amplifier configuration.

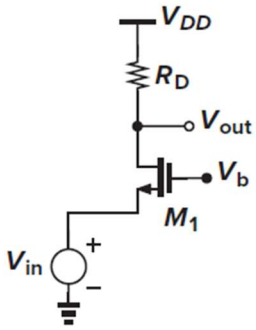


Figure 1. Single Stage CG Amplifier

Common drain amplifier is a very useful circuit for analog chips. Figure 2 show the CD configuration. The resistive load can be replaced with current source load using PMOS. The analysis of the NMOS current source-based load CD amplifier can be carried out by replacing the NMOS by its output resistance.

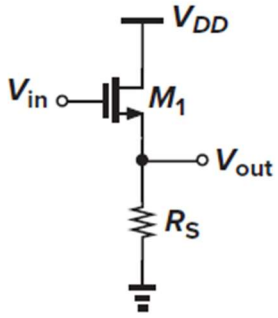


Figure 2. Single Stage CD Amplifier

Procedure (Attach a separate sheet with all calculations):

1. Refer to the values found in experiment 4 for resistive load CS amplifier. Same amplifier will be converted to common gate stage for appropriate comparison,
2. Prepare the circuit for simulation as per the values found out.
3. Carry out AC, DC, Transient, Noise and Harmonic analysis for the amplifier.
4. Compare theoretical results with the experimental results.
5. Repeat all the steps for common drain stage.
6. Compare the results with CS stage results.

Results:

Amplifier Parameters		
	CG	**CD**
Gain		
Bandwidth		
Noise		
Power		
Input AC voltage range		
Output AC voltage range		
MOSFET Parameters		
W driver		
L		
gm		
Ron		
μ_n		
ro		
V_{T0}		
C_{GS}		
I_{DS}		
C_{OX}		

Assignment:

1. What is the maximum ac voltage that can be applied to the amplifier for proper output operation? Consider both CG and CD stages.
2. What is the maximum output swing that can be achieved? Write the upper and lower value of output swing. Consider both CG and CD stages.
3. Compare the performance of CS stage with that of the CG and CD stage.

Conclusion:

Roll No. **Date:**

Exp. 7. To Design and simulate MOS based cascode amplifiers for given specifications.

Learning Outcomes:

After completion of this experiment students will be able to:

1. Explain the concept of cascode amplifier.
2. Design cascode amplifier and compare with its counterparts.
3. To carryout Monte Carlo analysis for given circuit.

Pre-lab Exercise

1. Draw the small signal model for cascode and folded cascode amplifier.

Introduction

A MOSFET-based cascode amplifier employs two MOSFETs arranged to enhance performance characteristics like gain, bandwidth, and linearity. The lower MOSFET serves as the common-source stage, providing high gain, while the upper MOSFET operates in a common-gate configuration, increasing output impedance and minimizing the Miller effect. This setup effectively reduces distortion and enhances stability, making it suitable for high-frequency applications such as RF amplifiers. The cascode configuration is particularly valued in integrated circuits for its ability to improve overall circuit performance while occupying a compact layout.

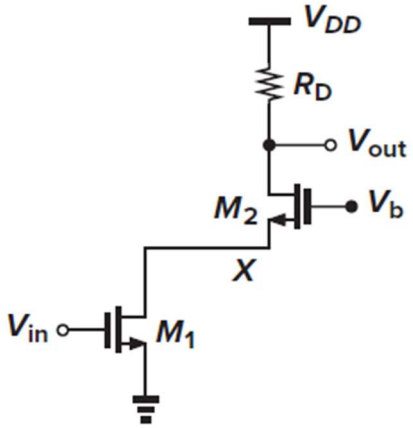


Figure 1. Cascode Amplifier

Procedure (Attach a separate sheet with all calculations):

1. Refer to the values found in experiment 4 resistive load CS amplifier. Same amplifier will be converted to cascode amplifier for appropriate comparison,
2. Prepare the circuit for simulation as per the values found out.
3. Carry out AC, DC, Transient, Noise and Harmonic analysis for the amplifier.
4. Compare theoretical results with the experimental results.
5. Compare the results with CS and CG stage results.

Results:

Amplifier Parameters			
Gain		Power	
Bandwidth		1 – dB Compression Point	
Noise		IIP3	
MOSFET Parameters			
L		Ro	
W		V_{T0}	
Gm		C_{GS}	
Ron		I_{DS}	
μ_n		C_{OX}	

Assignment:

1. List the advantage of folded cascode over cascode amplifier.
2. List the disadvantages of cascode amplifier compared to CS and CG amplifiers.

Conclusion:

Roll No. **Date:**

Exp. 8. To Design and simulate differential amplifiers for given specifications.

Learning Outcomes:

After completion of this experiment students will be able to:

1. Design and simulate differential amplifier for given specifications.
2. Carryout appropriate analysis for differential amplifier.
3. Evaluate performance parameters of differential amplifier.

Pre-lab Exercise

1. Explain the need of the tail current source in the differential amplifier.
2. List the applications, advantages and limitations of the differential amplifier.

Specifications:

V_{DD}=1.8V Gain = 30dB Power = 5 mW

Procedure:

1. Consider the PMOS load based differential amplifier for differential output.
2. For the given specifications find out the W for all the transistors involved in amplifier. (Use the concept of PMOS load CS amplifier)
3. Simulate the amplifier and find out the followings as per the observation table.
4. Convert the differential output to the single ended output using PMOS based Current Mirror load and observed the behaviour of the circuit. s

Observation Table:

Sr No	Parameters	Theoretical Value	Practical Value
1.	Differential Gain		
2.	Common mode Gain		
3.	I_{CMR}+		
4.	I_{CMR}-		
5.	Power dissipation		
6.	GBP		

Conclusion:

Roll No. **Date:**

Exp. 9. To design and simulate CMOS two stage operational Amplifier.

Learning Outcomes:

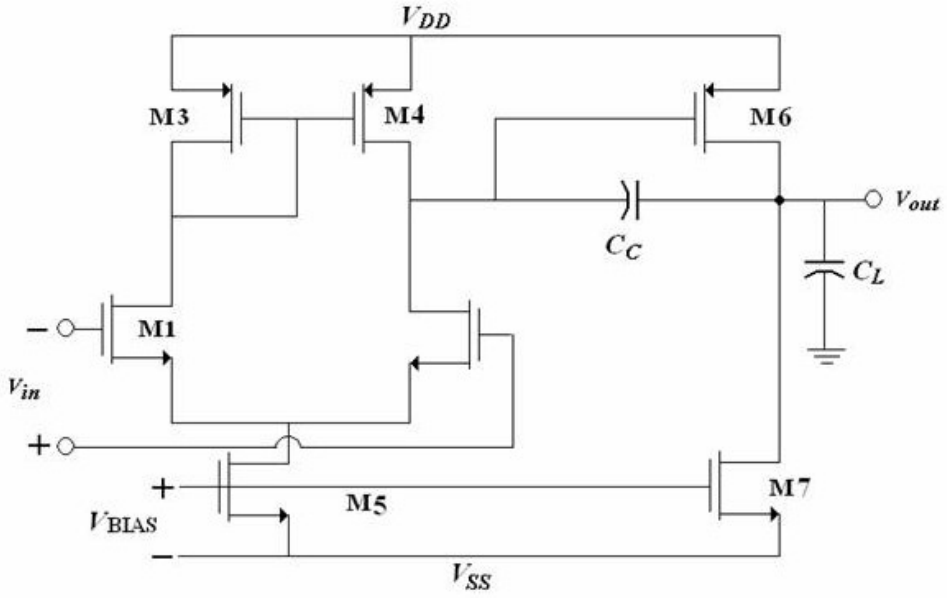
After completion of this experiment students will be able to:

1. Comprehend the concept of two stage operational amplifier.
2. Design and simulate operational amplifier.
3. Carry out analysis for and evaluate the performance of operational amplifier.
4. Utilize Opamp for various applications.
5. Prepare layout of Opamp and carry out post layout simulations.

Specifications:

V_{DD}=-V_{SS}=1.8V
Slew rate=10V/us
Gain = 200
CL=20pF
ICMR+=1.6V
ICMR-=0.8V
Power dissipation < 0.3mW
GBP=10MHz

Circuit Diagram:



Procedure:

1. Find out W/L for all the transistors for two stage opamp with given specifications for the circuit shown in figure above.
2. Prepare the schematic for simulation
3. Carry out necessary analysis to verify the specifications.
4. Compare the simulated result with theoretical calculations.
5. Prepare layout of designed circuit and carry out post layout simulations.

Assignment:

1. Derive the equation for DC gain and small signal gain of Opamp.
2. Elaborate the concept of miller capacitance. Discuss the effect of the same in the design of Opamp.
3. List few applications where Opamp can be employed. Verify any one of them using simulations.

Conclusion: